



Department of Electrical and Electronics Engineering
Academic Year 2024 – 2025 (Odd Semester)

Degree, Semester & Branch: VII Semester B.E. EEE

Course Code & Title: OCS 352 IoT Concepts and Applications

Name of the Faculty member: Mrs. G. Sivapriya, AP/EEE

Innovative Practice Description

- **Unit / Topic:** Unit I / IoT Architectures
- **Course Outcome:** CO 1
- **Topic Learning Outcome:** 1b
- **Activity Chosen:** Mind - Map
- **Justification:**
 - Explain the different types of IoT Architectures
 - After teaching the concept, I thought of conducting this activity for making the students to give the difference between the various types of IoT Architectures which enhance the learning level and as a teacher I can judge the understanding level of the students.

Time Allotted for the Activity: 6 Minutes

After teaching the concept, the students were made to pair with their neighbors

Photographer: Myself

Reporter: Myself

At the end the Class (Last 6 minutes)

- ✓ I asked the students to think about types of IoT Architectures concept for 2 minute.
- ✓ Then I told them to Pair with their neighbors and discuss about the concepts for another 1 minute.
- ✓ Finally I told them to design mind-map and submit within 3 minutes.

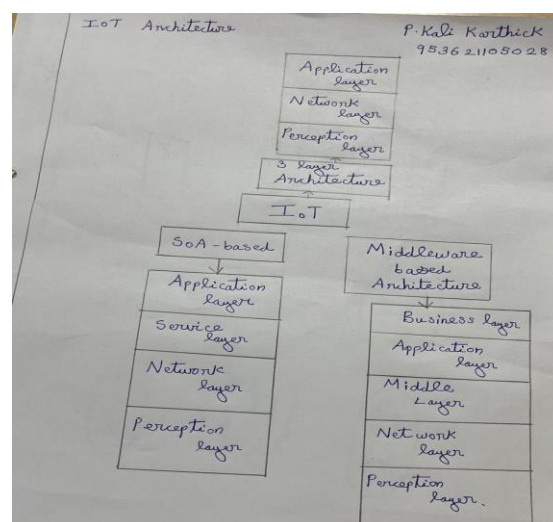
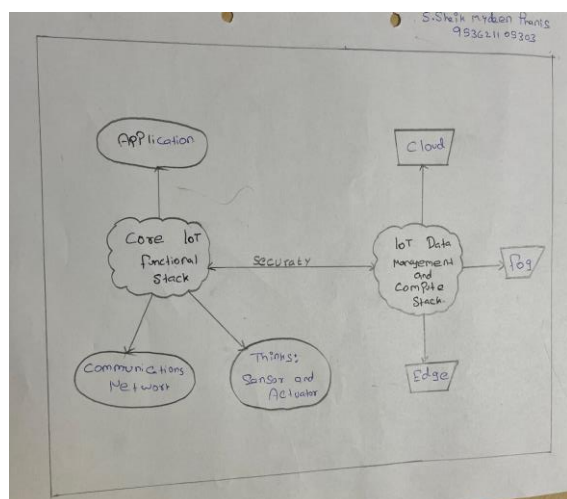
• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO9	PO10	PO12	PSO1
C404.1	2	2	3	2	1	1	1	1	1	1

- **PO / PSO mapped:**

Innovative practice	PO9	PO10
	1	1
Justification for correlation	The students are working in a group to list out the points and design a mind map. Hence it is mapped with level 2.	The students are drawing a neat mind map and trying to incorporate as much as points possible. Hence their visualization skills and presentation skills are improved. It is mapped with level 2.

- **Images / Screenshot of the practice:**



Reflective Critique:

1. **Pre-implementation Reflection:**

I preferred this Activity because they types of IoT Architectures separately.

Challenges anticipated:

- In the class few boys hesitate to answer to the questions.
- Time utilization for conducting activity.

Steps taken:

- The boys are sitting in 2 columns – I planned to choose more pairs from boys to involve them in the activity.

2. Post implementation Reflection:

Benefits:

- Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

Challenges:

- Slow learners were not able to understand some topics during discussion ~~has~~

❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- ✓ The assessment of effectiveness of the activity was felt when told most of the points.
- ✓ While conducting the activity, I understood that the students will be able to explain the concepts of types of Energy Storage Technologies.
- ✓ The success of the activity was evaluated by asking the same question in Internal Assessment test II – Around 90% of students answered.

References:

1. <https://omerad.msu.edu/teaching/teaching-strategies/active-learning-strategies/27-teaching/184-visual-modeling-mini-maps>

Signature of Faculty Member

HOD



Department of Electrical Electronics Engineering Academic Year 2024 – 2025 (Odd Semester)

Degree, Semester & Branch: B.E/V/EEE

Course Code & Title: OCS352/IoT Concepts and Applications

Name of the Faculty member (s): Mrs G.Sivapriya

Innovative Practice Description

- **Unit / Topic:** Unit II / Functional Blocks of an IoT Ecosystem
- **Course Outcome:** CO 2
- **Topic Learning Outcome:** 2a
- **Activity Chosen:** One Minute Paper
- **Justification:**

The functional blocks of an IoT Ecosystem have been discussed. After teaching this topic, I thought of this activity for making to have clarity about the Functional Blocks of an IoT Ecosystem. I can judge the understanding level of the students.

Time Allotted for the Activity: 5 Minutes

- **Details of the Implementation:**

After teaching the concept, I gave students one or two minutes to think about the topic without writing anything.

Total Strength :64

Reporter: Myself

At the end of the Class (Last 5 minutes)

I asked the students to think Functional Blocks of an IoT Ecosystem concept for 3 minutes.

Then I told them to write as much as they can within a short period of time (1 minute)

Finally, I collected the papers from each column

- **CO – PO / PSO mapping:**

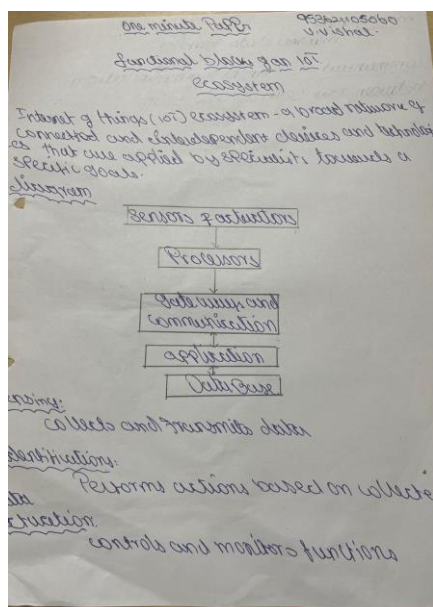
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO11	PO12	PSO1	PSO2
CO2	2	2	2	2	1	1	2	2	1	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO9
	1
Justification for correlation	The students can function effectively as an individual.

- **Images / Screenshot of the practice:**



one minute paper

Functional blocks of the IoT Ecosystem

The internet of things (IoT) ecosystem is a network of interconnected devices that communicate and exchange data to enable smart functionalities. The key functional blocks in a IoT ecosystem include:

Devices: These are the physical objects that collect or generate data and interact with their environment.

Connectivity: This layer ensures seamless communication between devices, often through wireless technologies like Wi-Fi, Bluetooth, Zigbee.

Edge computing: It refers to processing data closer to the source to reduce latency and bandwidth usage.

Data processing: Once data is collected and transmitted, this layer involves storing, processing and analyzing.

- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

- The assessment of effectiveness of the activity was felt when told most of the points.
 - While conducting the activity, I understood that the students will be able to explain Functional Blocks of an IoT Ecosystem.
 - The success of the activity was evaluated by asking the same question in Internal Assessment test I – Around 92% of students answered.

- Benefit of the practice:**

- Students were able to attend the question even in the questions are in indirect form.
 - Students were able to explain the concepts in examination without any confusion.

- ❖ **Challenges faced in implementation:**

- In the class mostly few students hesitate to answer the questions.
 - Time utilization for conducting activity.

References:

1. Ibrahim Dincer and Mark A. Rosen, 'Thermal Energy Storage Systems and Applications', John Wiley & Sons, 3rd Edition, 2021.
2. Ru-shi Liu, Lei Zhang and Xueliang sun, 'Electrochemical technologies for energy storage and conversion', Wiley publications, 2nd Volume set, 2012.
3. James Larminie and Andrew Dicks, 'Fuel cell systems Explained', Wiley publications, 3rd Edition, 2018.

Signature of Faculty Member

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Department of Electrical and Electronics Engineering
Academic Year 2024 – 2025 (Odd Semester)

Degree, Semester & Branch: VII Semester B.E. EEE

Course Code & Title: OCS 352 IoT Concepts and Applications

Name of the Faculty member: Mrs. G. Sivapriya, AP/EEE

Innovative Practice Description

- **Unit / Topic:** Unit III /IoT Protocols
- **Course Outcome:** CO3
- **Topic Learning Outcome:** 3a
- **Activity Chosen:** Flipped Classroom
- **Justification:**
 - Flipped classroom is a “pedagogical approach in which direct instruction moves from the group learning space to the individual learning space, and the resulting group space is transformed into a dynamic, interactive learning environment where the educator guides students as they apply concepts and engage creatively in the subject matter”

Time Allotted for the Activity: 30 Minutes

Number of Students –64(16 groups)

Photographer: one student – Mr. M. Suriya Prakash (interested in photography)

Reporter: Myself

At the Class (30 minutes)

- I shared the material related to the topic through canvas LMS.
- Then I told them to prepare the topic as a team with a help of PPT.

• **CO – PO / PSO mapping:**

3	3	2	2	1	1	-	2	-	-	-	2	2	2	-
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CO	PO1	PO2	PO3	PO5	PO8	PO9	PO10	PO12	PSO1	PSO2
C215.4	3	3	2	2	1	2	2	2	2	2

- **PO / PSO mapped:**

Innovative Practice	PO9	PO10
	2	2
Justification for correlation	Students will be able to work with their team through Active Learning Methods. Hence it is mapped with level 2.	Students will be able to communicate the technical concept through Active Learning Methods. Hence it is mapped with level 2.

• **Images / Screenshot of the practice:**



❖ *Reflective Critique:*

Implementation Reflection:

- **Challenges anticipated:**
 - Time utilization for conducting activity.
- **Benefits:**
 - This practice improves the peer coaching / studying with in the class.
 - Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

References:

1. <https://www.panopto.com/blog/7-unique-flipped-classroom-models-right/>

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Department of Electrical and Electronics Engineering
Academic Year 2024 – 2025 (Odd Semester)

Degree, Semester & Branch: VII Semester B.E. EEE

Course Code & Title: OCS 352 IoT Concepts and Applications

Name of the Faculty member: Mrs. G. Sivapriya, AP/EEE

Innovative Practice Description

- **Unit / Topic:** Unit V / IoT Applications-Home Automation
- **Course Outcome:** CO5
- **Topic Learning Outcome:** 5b
- **Activity Chosen:** Jigsaw
- **Justification:**
 - Jigsaw is a cooperative group activity in which students are interdependent to achieve a common goal. In part one, each group is provided a different prompt. The group members become experts on that prompt and create a group response. In part two, new groups are formed; comprised of students from different expert groups. Each student in the intermixed group is expected to teach the other group members their prompt-response from their previous group, “expert group”. The intermixed groups then complete a new task. The success of the group depends on each individual and therefore prompts engagement from individual students.
 - Explain the different types of IoT Applications
 - After teaching the concept, I thought of conducting this activity for making the students to give their idea about given particular task from assigned group and also from newly joined group.

Time Allotted for the Activity: 16 Minutes

After teaching the concept, the students were made to different (11) groups with their neighbors.

Photographer: Myself

Reporter: Myself

- ✓ I assigned a topic that home automation to all the group of students and allowed to discuss with team members for searching answer for 5 minute.
- ✓ Then I formed new group with new members and asked them to share with their neighbors and discuss about the concepts for another 4 minutes.

- ✓ Finally, Students work together to complete a task that requires them to use all the information they learned for 4 minutes.
- ✓ I asked questions to the students and analyzed their capability individually for 3 minutes.

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO9	PO10	PO12	PSO1
C404.5	3	3	2	2	2	2	3	2	2	2

• **PO / PSO mapped:**

Innovative practice	PO9	PO10
	3	2
Justification for correlation	The students are working in a group to list out the points and design a mind map. Hence it is mapped with level 3.	The students are drawing a neat mind map and trying to incorporate as much as points possible. Hence their visualization skills and presentation skills are improved. It is mapped with level 2.

• **Images / Screenshot of the practice:**



• *Reflective Critique:*

1. Pre-implementation Reflection:

I preferred this Activity for IoT Applications(Home Automation) separately.

Challenges anticipated:

- In the class few boys hesitate to answer to the questions.
- Crowded classrooms can make it difficult for students to be effectively involved.
- It can be challenging to form balanced and diverse groups. Students may have different levels of prior knowledge, interest, and ability, which can lead to

misunderstandings.

- The Jigsaw model requires individual accountability and group cooperation. Students must be prepared, responsible, and respectful in their roles and tasks
- Time utilization for conducting activity.

Steps taken:

- I have taken more time to prepare students to learn how to work in groups.

2. Post implementation Reflection:

Benefits:

- Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

Challenges:

- Slow learners were not able to explain some topics during discussions.

❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- ✓ The assessment of effectiveness of the activity was felt when told most of the points.
- ✓ While conducting the activity, I understood that the students will be able to explain the concepts individually and also from groups. They enjoyed in learning.
- ✓ The success of the activity was evaluated by asking the same question in Internal Assessment test II – Around 90% of students answered.

References:

1. <https://www.bell-foundation.org.uk/resources/great-ideas/jigsaw-activities/>
2. <https://www.digitaltheatreplus.com/blog/how-to-use-jigsaw-activities-in-the-classroom>

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